

InventOR: A Prompt-Configured, No-Fine-Tuning LLM Workflow for Material-Level Inventory Control

Aris Dressino^{a,*}

^a*TUM School of Management, Technical University of Munich, Arcisstraße
21, Munich, 80333, Germany*

Abstract

We present *InventOR*, a prompt-configured large-language-model (LLM) workflow that generates material-level (r, Q) inventory policies from cutoff-bounded historical CSVs and inline parameter blocks through an enterprise LLM application, without project-specific model training or fine-tuning. Deterministic parsing and post-cutoff simulation evaluate each LLM artifact against an SAP-derived and an SAP-safety-lead-time-informed operations-research comparator on a common 346-pair cohort from a three-plant industrial dataset. All 365 eligible plant-material pairs yield scoreable, capacity-feasible artifacts after retry handling. Working-day simulation reports 77.52% demand-weighted aggregate fill and 97.94% mean material fill for the LLM-emitted arm. Runs 2 and 3 show repeated completion but policy-value variation. Run 1 lacks preserved deployed prompt and runtime metadata; the contribution is a transparent descriptive evaluation protocol with explicit boundaries and current-run provenance controls for prospective evaluation.

Keywords: Generative Artificial Intelligence, Large Language Models,

*Corresponding author. Telephone: +49 89 289 25000.

Email address: `aris.dressino@tum.de` (Aris Dressino)

1 **1. Introduction**

2 Supply chain planning operates under volatile demand, uncertain replen-
3 ishment, and disruption exposure [1]. Many material-control decisions still
4 depend on simple parameters: reorder points, safety stock, order quantities,
5 and review periods; these settings strongly affect service and inventory expo-
6 sure [2, 3]. Classical operations research offers rigorous tools for stochastic
7 and robust inventory decisions [4, 5, 6], but planners often need a narrower
8 capability: converting imperfect historical records — which are pervasive in
9 practice [7] — into transparent, defensible, auditable material-level policy
10 values [8].

11 Large language models (LLMs) can help connect business intent, opera-
12 tional evidence, and analytical workflows, and autonomous decision-making
13 is an emerging capability frontier for supply chain management [9, 10, 11].
14 Recommendations in inventory control require explainability, reproducibil-
15 ity, and meaningful human control before planners can act on them [12, 13].
16 This creates a design requirement for hybrid AI-supported OR: LLM ar-
17 tifacts must be inspectable, and numeric recommendations require explicit
18 simulation and release controls [11, 13].

19 This paper addresses that requirement through *InventOR*, an operations-
20 research-first workflow for material-level inventory decision support. An en-
21 terprise LLM application produces policy artifacts from material-scoped in-
22 puts; we parse and simulate their numeric values deterministically.

23 Our contribution is an evaluation protocol rather than a new inventory

24 model or solution algorithm. The comparator policies are standard textbook
25 constructions; what we add is a reusable, auditable procedure for deciding
26 whether LLM-emitted policy values are admissible and how they perform
27 against ERP-derived baselines on a common simulation horizon. Three com-
28 ponents make up that protocol.

29 **C1 Admissibility protocol for LLM-emitted policy artifacts.**

30 Material-specific inputs pass through a configured enterprise LLM ap-
31 plication without project-specific training or fine-tuning. Deterministic
32 scoreability and capacity checks decide admission; the architecture
33 reserves a planner release step, which we document as a design
34 requirement rather than an evaluated intervention [14].

35 **C2 Executable boundary specification.** We document the material-
36 input contract, alias-aware conversion, capacity validation, and simula-
37 tion boundary, so that a competing generator can be substituted and
38 scored under identical rules.

39 **C3 Complete-cohort reporting discipline.** All 365 eligible pairs from
40 233,078 parsed records yielded scoreable, capacity-feasible artifacts af-
41 ter retry handling; 346 pairs meet the common feasibility rules used for
42 the three-arm descriptive comparison. No pair is silently dropped, and
43 every exclusion rule is reported.

44 The remainder of the paper is organised as follows. Section 2 reviews
45 related work and positions InventOR relative to recent AI-supported OR
46 contributions. Section 3 details the workflow, policy logic, and backtest
47 design. Section 4 reports completion and simulation results. Section 5 in-
48 terprets the findings and their limitations, and Section 6 summarises the

49 contribution and outlines future work.

50 **2. Literature Review**

51 Operations research has long addressed supply-chain uncertainty through
52 stochastic programming, robust optimization, and scenario-based planning
53 [4, 5, 6]. These methods formalize variability and recourse, but operational
54 inventory teams often face a more direct implementation problem: main-
55 taining reorder points, safety stock, safety lead time, and order quantities
56 from incomplete historical evidence in ERP-driven planning environments
57 [15, 16, 17, 18, 2, 3]. Static rules can be too coarse, while sophisticated mod-
58 els can be difficult to explain and maintain. This motivates decision support
59 that keeps OR discipline while making material-level parameter setting more
60 transparent and auditable [8, 19].

61 Recent LLM research proposes natural-language interfaces, tool-using
62 agents, and pipelines that translate natural-language requests into optimiza-
63 tion model structures or executable code, including multi-agent systems that
64 make autonomous per-period inventory decisions or seek consensus on opera-
65 tional actions [11, 20, 21, 22, 23, 24, 25, 26]. These approaches do not specify
66 how inventory histories should be reduced into demand regimes, lead-time
67 diagnostics, service buffers, and feasible policy parameters with deterministic
68 auditability.

69 We follow this hybrid view. We supply a filtered material CSV plus an
70 inline parameter block through an enterprise LLM application. The intended
71 prompt prohibits stochastic simulation, machine-learning libraries, and un-
72 restricted data fabrication, but stored run metadata do not independently
73 identify the deployed prompt or runtime. Because LLM outputs can contain

74 fabricated or unfaithful content [27], our parser uses alias-aware lookups and
75 nested-field resolution before requiring numeric positive reorder point and
76 order quantity, numeric non-negative safety stock, and a resolvable simula-
77 tor family. No project-specific model fitting or fine-tuning is performed. We
78 treat the LLM as an artifact-generation component inside a bounded decision
79 process. Table 1 positions our framework relative to recent AI-supported OR
80 and inventory-control work.

Table 1: Literature positioning: InventOR relative to recent AI-supported OR contribu-
tions for inventory and supply chain decision support.

Study	LLM role	Policy class	Demand model	Analytical result	Evaluation
Huang et al. [20]	NL→code	General OR	Not fied	speci- None	Benchmark
Ahmaditeshnizi et al. [22]	NL→model	General OR	Not fied	speci- Solver-linked	Benchmark
Zhang et al. [23]	Reasoning agent	General OR	Not fied	speci- Model solve	and Case studies
Quan and Liu [24]	Multi-agent control	Inventory	Sequential demand	Per-period de- cisions	Simulation
Prak et al. [28]	None	Inventory control	Compound Poisson / intermittent	Robust esti- mation	Empirical study
Babai et al. [29]	None	Forecasting-linked inventory	Regime-specific	Review thesis	syn- Review
Wasserkrug et al. [11]	Interface	General	Not fied	speci- None	Conceptual
Cohen et al. [21]	Vision	General SCM	Not fied	speci- None	Vision statement
InventOR	Prototype	(r, Q) executed; (s, S) im- plemented, unexer- cised	Artifact- defined values	Descriptive	Post-cutoff simulation

81 The table highlights the paper’s narrower contribution: material-level pa-
82 rameter artifacts evaluated by a deterministic policy-family simulator. Un-
83 like InvAgent’s sequential multi-agent decisions, InventOR emits one station-
84 ary policy per material from a bounded historical package; unlike OptiMUS
85 and OR-LLM-Agent, it does not formulate or solve general optimization

models. The (r, Q) framework is the workhorse of the continuous-review inventory literature [15, 18]; the intermittent-demand stream represented by Prak et al. [28] builds on Croston [30], Syntetos-Boylan [31], and Willemain et al. [32] to handle sporadic, non-normal demand patterns which our current prototype does not explicitly model.

3. Methodology

The unit of analysis is a plant-material pair. The framework records a chain of custody from historical records to planner-facing policy values by separating deterministic input preparation, configured LLM inference, schema parsing, deterministic simulation, and human review [8, 33]. Each LLM session receives one filtered, material-scoped historical CSV as a session resource and a small inline parameter block containing current cost, MOQ, service, SAP reference, and material-level storage-capacity fields. No project-specific model weights are trained, fine-tuned, or updated. The historical execution lacks frozen prompt, application-version, and model/runtime records, so it is not independently reproducible. Two later current-runner reruns record those provenance fields but form a distinct post-hoc execution set. Supplementary File S1 reproduces the intended application-side system and tool prompt texts verbatim, together with their disclosed discrepancies against the executable simulator.

3.1. Problem Definition: Single-Item Continuous-Review Control with Stress-Adjusted Service Buffering

The executed task is a single-item, single-echelon continuous-review simulation with lost sales [16, 17, 34]. Each plant-material pair is treated independently; multi-item coupling, shared capacity, substitution, and production

111 scheduling are outside the pilot. For the SAP-SLT-informed OR comparator,
 112 pre-cutoff working-day demand yields mean demand $\bar{d}_i$, annualized demand
 113 $\bar{D}_i^{\text{ann}} = 250\bar{d}_i$, lead-time-demand dispersion, and stationary (R_i, Q_i) param-
 114 eters. This comparator is not independent of ERP inputs because source
 115 SAP safety lead time enters effective lead time. For the SAP-derived arm,
 116 source safety stock is retained while R_i and Q_i are recomputed by the same
 117 executable routines. For the LLM arm, scoreable policy parameters origi-
 118 nate directly in the artifact and pass alias-aware compatibility parsing plus
 119 strict executable-contract validation; the reference equations below do not
 120 reconstruct them. The simulator executes validated (r, Q) artifacts as fixed-
 121 quantity policies and validated (s, S) artifacts as order-up-to policies. In this
 122 study all 365 scoreable artifacts resolved to the (r, Q) family after validation,
 123 so the order-up-to path is an available but unexercised capability rather than
 124 an evaluated result.

125 *Reference policy calculation.* The following equations describe the executed
 126 SAP-SLT-informed OR comparator; they do not reconstruct direct LLM-
 127 emitted policies. The order quantity $Q_i > 0$ is rounded upward to an integer,
 128 floored at source MOQ, and then projected to its material-level storage limit
 129 when possible. The reorder point $R_i \geq 0$ triggers replenishment. The EOQ
 130 anchor minimizes the standard annual cost approximation

$$\min_{Q_i} TC_i(Q_i; SS_i) = h_i \left(\frac{Q_i}{2} + SS_i \right) + K_i \frac{\bar{D}_i^{\text{ann}}}{Q_i},$$

131 where SS_i is arm-specific safety stock, $h_i = \text{holding_cost_rate}_i \cdot$
 132 unit_cost_i , and $K_i = \text{ordering_cost}_i$. The implementation first uses
 133 $Q_i = \max\{\text{MOQ}_i, \lceil \sqrt{2K_i(250\bar{d}_i)/h_i} \rceil\}$ when demand and ordering cost are
 134 positive, and MOQ otherwise. For every arm, a supplied capacity requires

135 $\text{MOQ}_i \leq Q_i$ and $SS_i + Q_i \leq \text{max_storage_units}_i$; comparator quantities
 136 above the residual limit are reduced, while pairs with $SS_i + \text{MOQ}_i$ above
 137 the limit are excluded from the common cohort. The SAP-SLT-informed
 138 OR safety stock is $SS_i = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil$, with $R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i \rceil$. The
 139 SAP-derived arm substitutes source SAP safety stock into the same reorder-
 140 point expression. No lot-size multiple or stress coefficient is executed. The
 141 primary holding/order cost has a zero shortage penalty; positive lost-sales
 142 penalties are sensitivity analyses.

143 *Core constraints.* Inventory evolves under a lost-sales continuous-review rule
 144 on the working-day decision set $\mathcal{W}$. All arms share the first observed post-
 145 cutoff on-hand inventory; the SAP-SLT-informed OR $R_i + Q_i$ is used only
 146 when that observation is unavailable. The initial on-order pipeline is empty.
 147 Duplicate same-material, same-date rows are aggregated by summing de-
 148 mand, receipts, corrections, and scrap and averaging positive lead times, but
 149 the simulator consumes only demand and excludes observed receipts, correc-
 150 tions, and scrap. The simulator rounds mean source lead time and schedules
 151 every modeled order by that number of subsequent working observations:

$$I_{i,t} = \max(0, I_{i,t-1} + a_{i,t} - d_{i,t}), \quad t \in \mathcal{W},$$

152 where $a_{i,t}$ is the quantity whose working-day due observation is no later than
 153 t . Scheduled arrivals are added first, realized demand is served subject to
 154 lost sales, and the post-demand inventory position is computed as on-hand
 155 plus outstanding orders. For a validated (r, Q) artifact, a fixed quantity is

156 ordered when

$$\delta_{i,t} = \mathbf{1}\{\text{IP}_{i,t}^+ \leq R_i\}, \quad \text{IP}_{i,t}^+ = I_{i,t} + \sum_{s: \text{due}(s) > t} o_{i,s},$$

157 with $o_{i,t} = Q_i \delta_{i,t}$. For a validated (s, S) artifact, the same threshold test uses
 158 s_i , and an order of $S_i - \text{IP}_{i,t}^+$ is placed when positive; no artifact in this study
 159 triggered that branch. Orders due after the observed horizon remain in tran-
 160 sit rather than being forced onto the final observation. Parameters remain
 161 stationary. Shared observed initialization and the empty initial pipeline are
 162 part of the executed comparison design.

163 The reorder point targets a nominal cycle service level $\alpha_i \in [0.5, 0.999]$
 164 after numeric clamping. Under the normal approximation,

$$\mathbb{P}(D_{i,L_i^{\text{eff}}} \leq R_i) \geq \alpha_i \iff R_i \geq \bar{d}_i L_i^{\text{eff}} + z_{\alpha_i} \sigma_{LTD,i}.$$

165 The reported backtest metric is the ex-post fill rate, which is distinct from
 166 the nominal cycle-service target because it also depends on Q_i and realized
 167 demand [35].

168 Lead-time-demand variance is propagated under the demand and lead-
 169 time independence approximation,

$$\sigma_{LTD,i}^2 = L_i^{\text{eff}} \sigma_{D,i}^2 + \bar{d}_i^2 \sigma_{L,i}^2.$$

170 The executed SAP-SLT-informed OR buffer is

$$SS_i^{\text{OR}} = z_{\alpha_i} \sigma_{LTD,i},$$

171 with values rounded upward:

$$SS_i^{\text{OR}} = \lceil z_{\alpha_i} \sigma_{LTD,i} \rceil, \quad R_i = \lceil \bar{d}_i L_i^{\text{eff}} + SS_i^{\text{OR}} \rceil.$$

172 The executable statistics add non-negative source SAP safety-lead-time days
 173 to mean lead time when constructing comparator lead-time-demand disper-
 174 sion (Section 3.5 discusses this confound). LLM-emitted values remain ar-
 175 tifact values accepted by alias-aware parsing and strict executable-contract
 176 validation.

177 *Parser and review boundary.* Artifact policy labels determine simulator rout-
 178 ing after validation. The parser resolves compatibility aliases, then requires
 179 a valid (r, Q) tuple or explicit s and S values for an (s, S) label; it rejects
 180 malformed, infeasible, or internally inconsistent artifacts. It does not execute
 181 label-specific optimization or deterministic escalation. Because no artifact
 182 supplied explicit and internally consistent s and S values, every scored policy
 183 in this study was executed as (r, Q) ; the reported evidence therefore concerns
 184 fixed-quantity control only. Planner review remains required before any op-
 185 erational use.

186 3.2. Historical Input Schema and Canonical Analysis Table

187 The uploaded CSV contains only the subset of source-table columns listed
 188 in Table 2. Plant and material identifiers are carried by the run key and in-
 189 line block rather than repeated in the uploaded rows. The inline block passes
 190 cost, MOQ, service, SAP reference, and material-level storage-capacity vari-
 191 ables. Stock-flow fields omitted from the uploaded subset are not analyzed
 192 by the executed LLM path.

Table 2: Fields included in the material-scoped CSV uploaded by the executed runner.

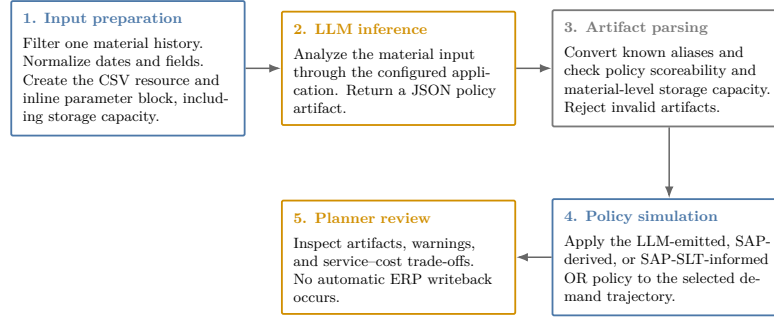
Column	Type	Description and method use
<code>date</code>	date / datetime	Calendar observation date; time ordering, aggregation, and diagnostics
<code>actual_demand</code>	numeric	Realized demand; mean, variability, intermittency, trend, and regime classification
<code>delivery_time</code>	numeric	Recorded replenishment lead time; rounded and scheduled as subsequent working observations by the simulator
<code>minimum_order_quantity</code>	numeric	Minimum replenishment quantity; order-size feasibility constraint
<code>safety_stock_units</code>	numeric	Existing safety stock; benchmarking against computed recommendations
<code>safety_time_</code> <code>workdays_actual</code>	numeric	Source SAP safety-lead-time field; currently enters comparator effective lead time
<code>supplier_plant_</code> <code>distance</code>	numeric	Supplier-to-plant distance; contextual lead-time-risk information
<code>reliability_calc</code>	numeric	Source supplier or lane reliability context
<code>transportation_network</code>	string	Source transportation-network context
<code>service_level_target</code>	numeric	Source service target used by comparator calculations
<code>unit_cost</code>	numeric	Purchase or standard cost per unit in the source dataset's internal cost denomination; inventory-value proxy and input to holding-cost derivation and EOQ calculation
<code>holding_cost_rate</code>	numeric	Annual carrying-cost rate as a decimal (e.g., 0.20 for 20% of unit value); used to derive the annual cost of carrying one unit in stock, balance replenishment frequency against stock exposure, and compute economic order quantity
<code>ordering_cost</code>	numeric	Fixed replenishment cost in the source dataset's internal cost denomination; input to EOQ calculation and simulated material-horizon holding/order cost
<code>max_storage_units</code>	numeric	Material-level upper bound applied to all evaluated arms: $MOQ \leq Q$ and $SS + Q \leq \text{max_storage_units}$
<code>is_working_day</code>	binary	Working-day flag; working-day demand statistics and non-zero demand share

193 3.3. Analytical Workflow

194 The workflow transforms a material snapshot into a policy artifact
195 through preparation, LLM inference, parsing, and simulation (Figure 1).

InventOR prototype workflow

Executed material-level path from prepared history to planner-reviewable simulation output.



Deterministic components reproduce simulation results from stored artifacts; LLM inference itself may vary.

Figure 1: Prototype workflow from material input preparation to policy artifact and simulation output.

196 The runner uploads a normalized CSV and inline block; the application
 197 returns a JSON artifact. The parser converts it to an executable (r, Q)
 198 or (s, S) policy and rejects malformed outputs. Missing artifacts are re-
 199 tried. The simulator evaluates LLM, SAP-derived, and SAP-SLT-informed
 200 OR policies on the common horizon. An aggregate-only manifest hashes
 201 active-input, evaluation-code, and stored-artifact files; it records unavail-
 202 able provenance fields explicitly. Current runner code records these fields
 203 when the corresponding environment variables are supplied. The author
 204 retrospectively identifies the historical model as Anthropic Claude Sonnet
 205 4.6 behind the enterprise application with a code-interpreter-style Python
 206 tool; this is not machine-verified provenance. Stored inputs, artifacts, and
 207 code reproduce a simulation, not a fresh LLM completion; planners retain
 208 implementation authority.

209 3.4. Artifact Completeness and Future Readiness

210 The completeness audit asks whether each eligible pair has a scoreable
 211 artifact after retries; it does not estimate first-pass reliability, explanation

completeness, or deployment readiness. The empirical section is a retrospective audit of an already-executed workflow rather than a controlled experiment: the generation cohort was produced before the evaluation protocol was finalized, the deployed configuration was not frozen or instrumented in advance, and no treatment was assigned. The post-cutoff simulation uses artifacts generated exclusively from pre-cutoff inputs and is descriptive rather than a controlled prospective trial [36, 37]. A future evaluation must freeze the deployed configuration, model/runtime identifiers, access controls, and approval workflow.

3.5. Safety Lead Time Computation

Safety lead time (SLT) is an ancillary ERP-facing design field. The prototype specification proposes combining transport, supplier, and quality lead-time variability with a reliability-deficit premium [16, 3], but the present release does not document estimators for those components. In the executed comparator code, non-negative source SAP safety-lead-time days are added to mean lead time before computing both SAP-derived and OR-computed dispersion and reorder points (this coupling is also flagged in Section 3.1). The OR arm is therefore labelled SAP-SLT-informed, and no separate SLT effect is identified.

3.6. Post-Cutoff Policy Simulation

The confidential operational extract contains 233,078 parsed CSV records, 411 plant-material pairs, and three anonymized plants (Plant A, Plant B, and Plant C). The parameter cutoff is 2026-03-10 and the post-cutoff simulation horizon ends on 2026-08-27. Eligibility filters retain 365 plant-material pairs with sufficient pre-cutoff and post-cutoff evidence:

237 234 from Plant A, 65 from Plant B, and 66 from Plant C. Plant C has
 238 no positive source working-day flags, so its calendar flags are derived
 239 from same-date Plant A and Plant B votes; ties classify as working days.
 240 This deterministic repair permits a common decision calendar; because
 241 no alternative Plant C calendar exists, robustness is examined by report-
 242 ing calendar-day arrivals and by re-running the full comparison on the
 243 native-calendar subset that excludes Plant C entirely. SAP-derived and
 244 SAP-SLT-informed OR parameters use pre-cutoff information. Stored LLM
 245 artifacts were generated from cutoff-bounded pre-cutoff inputs, so every
 246 arm is evaluated on the same post-cutoff horizon. Nineteen pairs cannot
 247 meet comparator safety stock plus MOQ within their supplied capacity;
 248 these are excluded before the common 346-pair comparison.

249 The simulated arms are: (i) SAP-derived, retaining source SAP safety
 250 stock while recomputing reorder point and EOQ order quantity from pre-
 251 cutoff statistics; (ii) SAP-SLT-informed OR, using the same statistics, source
 252 safety-lead-time input, and feasibility rules but without LLM inference; and
 253 (iii) LLM-emitted, a validated (r, Q) or (s, S) artifact. The simulator is
 254 continuous-review and lost-sales: all arms begin with the same first observed
 255 post-cutoff on-hand inventory, using the OR $ROP + Q$ only as a shared fall-
 256 back. The initial on-order pipeline is empty. On each working row, modeled
 257 arrivals due after rounded mean working-day lead time are received, demand
 258 is served subject to lost sales, post-demand inventory position is checked,
 259 and a policy-specific order is placed if its threshold is met. Observed post-
 260 cutoff receipts, corrections, and scrap are excluded. This common startup
 261 design may create cutoff transients. Metrics are fill rate, aggregate demand-
 262 weighted fill rate, materials above 95% fill, zero-stockout materials, stockout
 263 days, average on-hand inventory, and simulated material-horizon cost.

For policy p and material i , the cost summaries use the simulated horizon of T_i working days. Let $\bar{I}_{i,p}$ be average on-hand inventory, h_i annual holding cost per unit, K_i fixed ordering cost, and $n_{i,p}^{\text{orders}}$ simulated orders. The implementation uses a constant 250 annual working days and computes

$$TC_{i,p} = h_i \bar{I}_{i,p} \frac{T_i}{250} + K_i n_{i,p}^{\text{orders}}.$$

This is a simulated material-horizon holding/order cost in the source dataset’s internal cost denomination, not an audited annual financial or booked accounting result. Table 3 reports arithmetic means across the cohort with zero shortage penalty. Sensitivities add lost-sales penalties of one and five times source unit cost; cost values must be read jointly with fill rate and stockout days.

4. Results

4.1. Completion and Artifact Validity

The final execution produced parser-scoreable artifacts for all 365 eligible plant-material pairs with zero missing and zero parser-rejected outputs after retry handling. Every artifact passes alias-aware parsing, nested-field resolution, and capacity validation against the supplied material-level storage bound; all resolve to the (r, Q) family after validation.

4.2. Main Three-Arm Exploratory Simulation

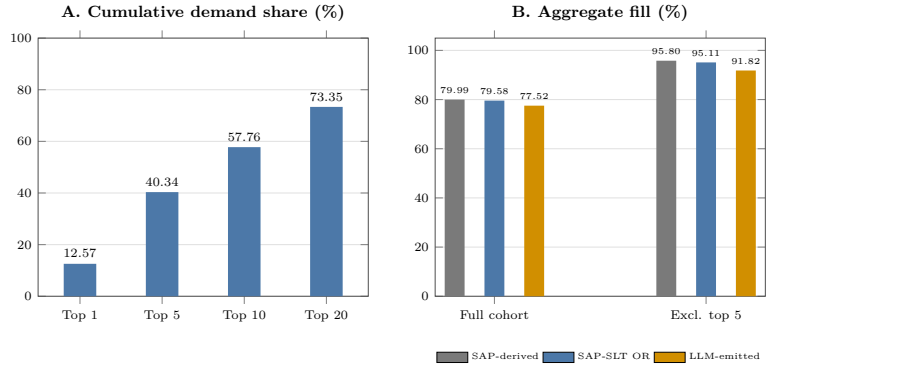
Table 3 reports three policy arms on the same 346 common-feasible material trajectories under working-day post-cutoff simulation. All arms satisfy identical MOQ and supplied per-material storage rules; 19 otherwise eligible pairs whose comparator safety stock plus MOQ exceed capacity are excluded.

Table 3: Post-cutoff working-day simulation on 346 common-feasible plant-material pairs. All arms use pre-cutoff parameterization or cutoff-bounded inputs and satisfy identical MOQ and supplied storage rules. Cost is mean simulated material-horizon holding/order cost with zero shortage penalty. SO days is the total count of stockout days across the cohort.

Policy	<i>n</i>	Mean FR (%)	Agg. FR (%)	$\geq 95\%$ (%)	Zero SO (%)	Mean SO cost	SO days
SAP-derived	346	98.38	79.99	94.51	80.35	3,184.20	741
SAP-SLT	346	98.27	79.58	93.93	88.73	3,852.22	799
OR							
LLM-emitted	346	97.94	77.52	92.49	81.50	2,848.04	936

Demand concentration and fill sensitivity

Largest-demand materials strongly influence full-cohort aggregate fill.



Generated from the working-days primary backtest; common-feasible cohort $n=346$.

Figure 2: Demand concentration and aggregate-fill sensitivity. Panel A reports cumulative demand share for the largest materials. Panel B compares full-cohort aggregate fill with the value after excluding the five largest-demand materials.

286 Relative to SAP-derived, LLM-emitted policies show 2.48 percentage
287 points lower aggregate fill, 336.16 lower mean holding/order cost, and 195
288 additional stockout days. Relative to SAP-SLT-informed OR, 2.07 percent-
289 age points lower aggregate fill, 1,004.18 lower mean holding/order cost, and
290 137 additional stockout days.

291 Aggregate evidence is demand-concentrated. The largest 1, 5, 10, and
292 20 materials contribute 12.57%, 40.34%, 57.76%, and 73.35% of post-cutoff
293 demand. Excluding the five largest-demand materials raises aggregate fill
294 to 91.82% for LLM-emitted, 95.80% for SAP-derived, and 95.11% for SAP-

SLT-informed OR, versus 77.52–79.99% in the full cohort (Figure 2). The largest-demand items drive aggregate fill.

At material level, LLM-emitted fill is not lower than SAP-derived for 314 of 346 materials (90.8%), while its cost is no higher for 238 (68.8%); both conditions hold for 211 (61.0%). Against SAP-SLT-informed OR, the corresponding counts are 309 (89.3%), 304 (87.9%), and 270 (78.0%). Paired LLM-minus-comparator fill differences have first quartile, median, and third quartile of 0.00 percentage points for both comparators, reflecting many exact fill ties. Corresponding cost differences are -259.24 , -47.17 , and 34.96 versus SAP-derived and -787.83 , -209.32 , and -25.26 versus SAP-SLT-informed OR. A deterministic 10,000-resample paired bootstrap gives LLM-minus-SAP fill and cost means of -0.44 percentage points (95% interval $[-0.99, -0.02]$) and -336.16 (95% interval $[-630.53, -106.52]$); against SAP-SLT-informed OR, the corresponding values are -0.33 percentage points ($[-0.89, 0.14]$) and $-1,004.18$ ($[-1,363.81, -705.90]$). These descriptive resampling intervals and aggregate results show service–cost trade-offs rather than uniform policy ranking.

The artifact audit covers all 365 scoreable policies: normalized policy-family agreement is 91.51%, and 42 artifacts meet a reported diagnostic-outlier rule (CV absolute error above 0.75 or LLM-to-pipeline safety-stock ratio outside $[0.5, 2.0]$). Removing the 23 flagged artifacts present in the common cohort raises LLM aggregate fill from 77.52% to 79.55%, while SAP-derived reaches 80.08% and SAP-SLT-informed OR reaches 79.70%; this post-hoc diagnostic subset does not establish a causal correction. Replacing primary working-day receipt offsets with calendar-day offsets gives aggregate fills of 77.77%, 80.25%, and 79.84%, respectively. Removing the one plant whose source working-day flags are entirely absent, and whose calendar is

322 therefore imputed, leaves 280 native-calendar pairs with aggregate fills of
 323 74.03% for LLM-emitted, 76.55% for SAP-derived, and 76.35% for SAP-
 324 SLT-informed OR, mean material-level fills of 98.19%, 98.71%, and 98.54%,
 325 zero-stockout shares of 81.79%, 88.21%, and 90.71%, and mean holding/order
 326 costs of 2,893.39, 3,471.66, and 4,168.76. In this native-calendar subset both
 327 comparators retain higher fill and higher zero-stockout shares than the LLM
 328 arm, so the imputed calendar does not manufacture the observed service
 329 gap. With a lost-sales penalty of one times unit cost, mean total costs
 330 are 1,688,921.83, 1,631,321.28, and 1,692,746.91, respectively; at five times
 331 unit cost, they are 8,433,217.00, 8,143,869.56, and 8,448,325.64. Thus, zero-
 332 penalty holding/order cost cannot establish economic superiority. These
 333 sensitivities do not resolve empty-pipeline, excluded-flow, or calendar-repair
 334 limitations.

335 *4.3. Three-Run Reproducibility Check*

336 Runs 1, 2, and 3 each produced 365 output artifacts. The common
 337 material cohort is defined as plant-material artifact identifiers with parseable
 338 policy triples in all three runs; it contains 365 pairs. All Figure 3 distributions
 339 and provenance counts use only this cohort. Runs 2 and 3 have 359 artifacts
 340 with complete matching provenance contracts; six are excluded because at
 341 least one required provenance field is absent. Among the 359 parseable policy
 342 pairs, all three numeric policy values match exactly for 38 pairs, resolved
 343 policy family matches for 357 pairs, and the median absolute differences
 344 are 122 units for reorder point, 5 units for order quantity, and 4 units for
 345 safety stock. Raw selected-model labels match for 12 pairs only, illustrating
 346 output-schema label variation rather than a second execution path. On
 347 the 338-pair outcome-common cohort, LLM aggregate fill is 79.05% in Run

348 2 and 79.09% in Run 3, mean material fill is 97.94% and 98.11%, mean
349 holding/order cost is 2,861.73 and 2,880.12, and stockout days total 918
350 and 846. The corresponding deterministic comparators are identical on that
351 aligned cohort. Run 1 predates provenance instrumentation; Runs 2 and 3
352 establish repeated completion under the current instrumented runner, but
353 policy differences and incomplete provenance prevent a claim of stable policy
354 values.

Three-run policy comparison											
Shared material cohort ($n = 365$). Policy cells report Q1 median Q3 units.											
Run	Artifacts	Complete provenance	Reorder point			Order quantity			Safety stock		
			Q1	median	Q3	Q1	median	Q3	Q1	median	Q3
Run 1	365	0	728	1,580	5,181	1,337	2,767	8,407	405	683	2,239
Run 2	365	364	784	1,507	4,793	1,341	2,689	8,653	404	688	2,177
Run 3	365	360	759	1,522	4,814	1,337	2,737	8,640	400	698	2,212

All values use the shared material cohort ($n=365$). Run 1 predates provenance instrumentation and has no complete frozen provenance. Runs 2-3: 359 provenance-matched policy pairs; 38 exact numeric triples.

Figure 3: Aggregate policy comparison across the 365-pair shared material cohort. Every row uses plant-material artifact identifiers with parseable policy triples in Runs 1, 2, and 3. Policy values report the first quartile, median, and third quartile. Run 1 predates provenance instrumentation; Runs 2 and 3 record current-runner provenance.

355 5. Discussion

356 We identify three findings as the principal scientific contribution. First,
357 100% of 365 eligible plant-material pairs yield scoreable, capacity-feasible
358 policy artifacts after retry handling; complete-cohort accounting avoids silent
359 dropout. Second, all scored artifacts resolve to the (r, Q) family; the LLM
360 constructs every policy value from only the material-scoped history and pa-
361 rameter block, while both comparators use ERP-derived inputs that em-
362 bed accumulated planner judgment. The observed aggregate-fill deficit of
363 roughly two percentage points and higher stockout-day count (936 versus
364 741 for SAP-derived and 799 for SAP-SLT-informed OR) are consistent
365 with this cold-start asymmetry, but do not identify its causal effect. An
366 ERP-independent cold-start OR arm is required to isolate generator quality.
367 Third, positive lost-sales penalty sensitivities remove the zero-penalty cost
368 ranking, so that ranking cannot establish generator quality.

369 The evaluation protocol is the reusable contribution. An aggregate-only
370 manifest hashes active inputs, evaluation code, and stored artifacts, flag-
371 ging unavailable provenance fields. The input-contract specification, alias-
372 aware parser, capacity-validation rules, and the full exclusion log form a
373 reproducible boundary within which any LLM or OR generator can be sub-
374 stituted and scored under identical rules. Runs 2 and 3 add evidence on
375 current-runner completion and policy variation relative to Run 1: six shared
376 artifacts lack complete provenance and only 38 of 359 provenance-matched
377 pairs have identical numeric policies. Future repeated-inference studies must
378 freeze the configuration before generation and define the alignment analysis
379 ex ante [38].

380 Our study is bounded by a single anonymized three-plant dataset, an

381 uninstrumented Run 1 configuration, two post-hoc current-runner executions
382 (Runs 2 and 3), and a post-cutoff simulation that excludes observed receipts,
383 corrections, and scrap with an empty initial pipeline. These are explicit
384 limitations. The contribution is the protocol and its descriptive execution,
385 not a deployment-performance claim.

386 6. Conclusion

387 We present *InventOR*, an applied AI/OR prototype that treats LLMs as
388 candidate-policy generators within deterministic preparation, parsing, and
389 simulation. The framework preserves an inspectable link between mate-
390 rial histories, policy fields, and simulated outcomes without project-specific
391 model training or fine-tuning.

392 The executed artifact set contains 365 scoreable policies generated from
393 cutoff-bounded, pre-cutoff material histories, all satisfying their material-
394 level storage bounds. On the 346-pair cohort where all arms satisfy identi-
395 cal MOQ and supplied storage rules, LLM-emitted aggregate fill is 77.52%,
396 mean material-level fill is 97.94%, mean simulated material-horizon hold-
397 ing/order cost is 2,848.04, and stockout days total 936 under working-day
398 receipt scheduling. The ERP-informed comparator construction prevents
399 attributing the service gap to generator quality. Positive shortage-cost sen-
400 sitivities remove the zero-penalty cost advantage. Runs 2 and 3 confirm
401 repeated completion but show policy-value variation. These values are de-
402 scriptive simulated-comparison evidence, not deployment or production-trial
403 evidence.

404 Future work should supply an ERP-independent cold-start OR arm,
405 predefine repeated-inference alignment and stability measures before

406 cohort-scale generation, extend the admissible policy family beyond (r, Q)
407 to order-up-to and intermittent-demand regimes, and run a prospective
408 evaluation with planner-in-the-loop review and fixed instrumentation. An
409 ex-ante diagnostic-outlier exclusion rule and a documented SLT-component
410 estimator belong in the same evaluation protocol upgrade.

411 **Author Contributions**

412 Conceptualization, Methodology, Software, Formal analysis, Validation,
413 Visualization, Writing – original draft, Writing – review and editing: Aris
414 Dressino.

415 **Funding**

416 This research did not receive any specific grant from funding agencies in
417 the public, commercial, or not-for-profit sectors.

418 **Declaration of Competing Interest**

419 The author declares no known competing financial interests or personal
420 relationships that could have appeared to influence the work reported in this
421 paper.

422 **Data and Code Availability**

423 The raw operational data and material-level artifacts cannot be pub-
424 licly released because they contain commercially sensitive material, supplier,

425 and planning information. A confidentiality-screened release package con-
426 taining analysis code, the anonymized variable schema, aggregate evalua-
427 tion results, manuscript sources, and aggregate integrity manifest is avail-
428 able at github.com/DressPD/inventor_public. The GitHub repository is
429 the development mirror. A versioned Zenodo archive will be created be-
430 fore archival release; no Zenodo DOI is claimed for this submission. Sup-
431 plementary File S1 accompanies this submission and contains the intended
432 application-side prompt specification. Run 1 deployed application configu-
433 ration, model/runtime identifier, inference dates, and prompt hashes were
434 not preserved and are not claimed as available; the current runner records
435 these fields for future executions.

436 **7. Acknowledgments**

437 The author thanks the industrial collaborators and material-control ex-
438 perts who contributed domain knowledge and access to anonymized oper-
439 ational data. The author also acknowledges the associated enterprise data
440 and IT teams for supporting data access and prototype execution. No for-
441 mal full-population planner evaluation is claimed. Any errors or omissions
442 remain the responsibility of the author.

443 **Declaration of Generative AI and AI-Assisted Technologies in the** 444 **Writing Process**

445 During manuscript preparation, the author used OpenAI GPT-5.6
446 through OpenCode for language editing, consistency checks, citation review,
447 and LaTeX proofing support; this is a writing-assistance tool and is distinct
448 from the Anthropic Claude Sonnet 4.6 deployment evaluated in the research

449 workflow. The author reviewed and edited the content after use and takes
450 full responsibility for the content of the published article. Figures were
451 rendered deterministically from author-reviewed LaTeX and Python source;
452 generative AI did not directly create or manipulate image content. The
453 research workflow evaluated in the paper is itself the object of study and is
454 described separately in the methodology.

455 References

- 456 [1] S. Sengupta, P. Jonsson, H. Dreyer, R. Kaipia, T. Y. Choi, Uncertainty regula-
457 tion and adaptable supply chain planning, *International Journal of Operations*
458 *& Production Management* 45 (11) (2025) 1861–1883.
- 459 [2] E. T. Grasso, B. W. Taylor, A simulation-based experimental investigation of
460 supply/timing uncertainty in MRP systems, *International Journal of Produc-*
461 *tion Research* 22 (3) (1984) 485–497. doi:10.1080/00207548408942468.
- 462 [3] T. J. van Kampen, D. P. van Donk, D.-J. van der Zee, Safety stock
463 or safety lead time: coping with unreliability in demand and supply,
464 *International Journal of Production Research* 48 (24) (2010) 7463–7481.
465 doi:10.1080/00207540903348346.
- 466 [4] D. Bertsimas, A. Thiele, A robust optimization approach to inventory theory,
467 *Operations Research* 54 (1) (2006) 150–168. doi:10.1287/opre.1050.0238.
- 468 [5] A. Thorsen, T. Yao, Robust inventory control under demand and lead time
469 uncertainty, *Annals of Operations Research* 257 (1) (2017) 207–236.
- 470 [6] D. Zhang, H. H. Turan, R. Sarker, D. Essam, Robust optimization approaches
471 in inventory management: Part A—the survey, *IIE Transactions* 57 (7) (2025)
472 818–844.
- 473 [7] N. DeHoratius, A. Raman, Inventory record inaccuracy: An empirical analysis,
474 *Management Science* 54 (4) (2008) 627–641. doi:10.1287/mnsc.1070.0789.

- 475 [8] A. J. Mersereau, Information-sensitive replenishment when inventory records
476 are inaccurate, *Production and Operations Management* 22 (4) (2013) 792–
477 810.
- 478 [9] T. Boone, B. Fahimnia, R. Ganeshan, D. M. Herold, N. R. Sanders, Gener-
479 ative AI: Opportunities, challenges, and research directions for supply chain
480 resilience, *Transportation Research Part E: Logistics and Transportation Re-*
481 *view* 199 (2025) 104135.
- 482 [10] I. Jackson, D. Ivanov, A. Dolgui, J. Namdar, Generative artificial intelligence
483 in supply chain and operations management: A capability-based framework
484 for analysis and implementation, *International Journal of Production Research*
485 62 (17) (2024) 6120–6145. doi:10.1080/00207543.2024.2309309.
- 486 [11] S. Wasserkrug, L. Boussioux, D. Den Hertog, F. Mirzazadeh, Ş. İ. Birbil,
487 J. Kurtz, D. Maragno, Enhancing decision making through the integration of
488 large language models and operations research optimization, in: *Proceedings*
489 *of the AAAI Conference on Artificial Intelligence*, Vol. 39, 2025, pp. 28643–
490 28650. doi:10.1609/aaai.v39i27.35090.
- 491 [12] S. Amershi, D. Weld, M. Vorvoreanu, A. Fournay, B. Nushi, P. Collisson,
492 J. Suh, S. Iqbal, P. N. Bennett, K. Inkpen, et al., Guidelines for human-AI
493 interaction, in: *Proceedings of the 2019 CHI Conference on Human Factors in*
494 *Computing Systems*, 2019, pp. 1–13.
- 495 [13] I. D. Raji, A. Smart, R. N. White, M. Mitchell, T. Gebru, B. Hutchinson,
496 J. Smith-Loud, D. Theron, P. Barnes, Closing the AI accountability gap:
497 Defining an end-to-end framework for internal algorithmic auditing, in: *Pro-*
498 *ceedings of the 2020 Conference on Fairness, Accountability, and Transparency*,
499 2020, pp. 33–44. doi:10.1145/3351095.3372873.
- 500 [14] P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, A. Chadha,
501 A systematic survey of prompt engineering in large language mod-

els: Techniques and applications, arXiv preprint arXiv:2402.07927 (2024).
doi:10.48550/arXiv.2402.07927.

[15] G. Hadley, T. M. Whitin, Analysis of Inventory Systems, Prentice-Hall, Englewood Cliffs, NJ, 1963.

[16] E. A. Silver, D. F. Pyke, D. J. Thomas, Inventory and Production Management in Supply Chains, 4th Edition, CRC Press, Boca Raton, FL, 2017.

[17] P. H. Zipkin, Foundations of Inventory Management, McGraw-Hill, New York, 2000.

[18] E. L. Porteus, Foundations of Stochastic Inventory Theory, Stanford University Press, Stanford, CA, 2002.

[19] Y. Kang, S. B. Gershwin, Information inaccuracy in inventory systems: stock loss and stockout, IIE transactions 37 (9) (2005) 843–859.

[20] C. Huang, Z. Tang, S. Hu, R. Jiang, X. Zheng, D. Ge, B. Wang, Z. Wang, ORLM: A customizable framework in training large models for automated optimization modeling, Operations Research 73 (6) (2025) 2986–3009. doi:10.1287/opre.2024.1233.

[21] M. C. Cohen, T. Dai, G. Perakis, N. Agrawal, G. Allon, R. N. Boute, G. P. Cachon, Z. Chen, M. Cohen, R. Cristian, et al., OM Forum—Supply Chain Management in the AI Era: A vision statement from the operations management community, Manufacturing & Service Operations Management 28 (3) (2026) 687–705. doi:10.1287/msom.2025.1065.

[22] A. Ahmaditeshnizi, W. Gao, M. Udell, OptiMUS: Scalable optimization modeling with (MI)LP solvers and large language models, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 577–596.
URL <https://proceedings.mlr.press/v235/ahmaditeshnizi24a.html>

- 528 [23] B. Zhang, P. Luo, G. Yang, B.-H. Soong, C. Yuen, OR-LLM-Agent:
529 Automating modeling and solving of operations research optimization
530 problems with reasoning LLM, arXiv preprint arXiv:2503.10009 (2025).
531 doi:10.48550/arXiv.2503.10009.
- 532 [24] Y. Quan, Z. Liu, InvAgent: A large language model based multi-agent system
533 for inventory management in supply chains, arXiv preprint arXiv:2407.11384
534 (2024). doi:10.48550/arXiv.2407.11384.
- 535 [25] B. Li, K. Mellou, B. Zhang, J. Pathuri, I. Menache, Large language mod-
536 els for supply chain optimization, arXiv preprint arXiv:2307.03875 (2023).
537 doi:10.48550/arXiv.2307.03875.
- 538 [26] R. Jannelli, M. Schöpf, L. Bickel, T. H. Netland, A. Brintrup,
539 Agentic LLMs in the supply chain: Towards autonomous multi-agent
540 consensus-seeking, International Journal of Production Research (2025).
541 doi:10.1080/00207543.2025.2604311.
- 542 [27] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto,
543 P. Fung, Survey of hallucination in natural language generation, ACM Com-
544 puting Surveys 55 (12) (2023) 1–38. doi:10.1145/3571730.
- 545 [28] D. Prak, R. Teunter, M. Z. Babai, J. E. Boylan, A. Syntetos, Robust compound
546 Poisson parameter estimation for inventory control, Omega 104 (2021) 102481.
547 doi:10.1016/j.omega.2021.102481.
- 548 [29] M. Z. Babai, A. A. Syntetos, R. H. Teunter, Fifty years of inventory research
549 from a forecasting perspective, European Journal of Operational Research
550 331 (1) (2026) 1–20. doi:10.1016/j.ejor.2025.07.003.
- 551 [30] J. D. Croston, Forecasting and stock control for intermittent demands, Oper-
552 ational Research Quarterly 23 (3) (1972) 289–303. doi:10.1057/jors.1972.50.

- 553 [31] A. A. Syntetos, J. E. Boylan, The accuracy of intermittent demand
554 estimates, *International Journal of Forecasting* 21 (2) (2005) 303–314.
555 doi:10.1016/j.ijforecast.2004.10.001.
- 556 [32] T. R. Willemain, C. N. Smart, H. F. Schwarz, A new approach to forecast-
557 ing intermittent demand for service parts inventories, *International Journal of*
558 *Forecasting* 20 (3) (2004) 375–387. doi:10.1016/S0169-2070(03)00013-X.
- 559 [33] S. Watts, G. Shankaranarayanan, A. Even, Data quality assessment in context:
560 A cognitive perspective, *Decision Support Systems* 48 (1) (2009) 202–211.
- 561 [34] M. Bijvank, I. F. A. Vis, Lost-sales inventory theory: A re-
562 view, *European Journal of Operational Research* 215 (1) (2011) 1–13.
563 doi:10.1016/j.ejor.2011.02.004.
- 564 [35] S. Axsäter, *Inventory Control*, 3rd Edition, Vol. 225 of International Series in
565 *Operations Research & Management Science*, Springer, Cham, 2015.
- 566 [36] L. J. Tashman, Out-of-sample tests of forecasting accuracy: an analysis
567 and review, *International Journal of Forecasting* 16 (4) (2000) 437–450.
568 doi:10.1016/S0169-2070(00)00065-0.
- 569 [37] H. Hewamalage, K. Ackermann, C. Bergmeir, Forecast evaluation for data
570 scientists: common pitfalls and best practices, *Data Mining and Knowledge*
571 *Discovery* 37 (2) (2023) 788–832. doi:10.1007/s10618-022-00894-5.
- 572 [38] S. Ouyang, J. M. Zhang, M. Harman, M. Wang, An empirical study of the non-
573 determinism of ChatGPT in code generation, *ACM Transactions on Software*
574 *Engineering and Methodology* 34 (2) (2025) 1–28. doi:10.1145/3697010.